

ForeShift Development Specification v7.1

Web and Bubble.io implementation guide — scoring logic, runtime, holiday/event layers, and data export

Model master: ForeShift_Detroit_v71.xlsx · 10 Detroit zones · 560 baseline rows · holiday/occasion layer added

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What changed in this revision (vs the prior v7.1 draft). Added the holiday / calendar-occasion layer (two new tables, a signed runtime term, and a floor at 0); an explicit rule that event_affinity must never be applied to holidays; a holiday+event stacking decision; the band edge-trend display rule; a new narration-layer section (§5.5); holiday acceptance tests; and a utilization reconciliation to 1.0. Sections not listed are unchanged.

1. Architecture principle: Excel is the model master, the app is the delivery layer

ForeShift separates heavy model construction (Phase A) from lightweight runtime delivery (Phase B). The web/Bubble application implements ONLY Phase B. All Phase A work — Location Scores, Concept Scores, capture-weight blends, the zone-daypart resolution, venue-supply evidence, the 560 baseline values, and the holiday affinity profiles — happens in the v71 Excel workbook and is exported as flat data. The app does lookups, one scoring formula, and narration.

Two layers, never merged. Deterministic scoring (this spec, §3) produces a number and a band. The narration layer (§5.5) turns that into operator-facing language. The narration layer **narrates, it never computes** — it may not alter scores, only describe them.

2. Data model

Ten core tables. Fields marked [TS] are trade secret: store server-side, never expose via API responses, page data, or client-visible searches. Operators see bands, statuses, and recommendations — never raw coefficients.

Table	Key fields	Rows	Notes
Zone	name, display_name, geo_center (lat/lng), geo_radius	10	Includes Mexicantown. Geo drives event-API queries.
ConceptType	name, daypart_allocation [TS], turn_rates [TS], event_affinity [TS], weather_affinity [TS]	8	event_affinity applies to EventMagnitude ONLY (see §3.1).
BaselineCurve	zone, concept_type, day_of_week, structural_base [TS], daypart_fraction (4) [TS]	560	Uncapped, seasonal-free. Sports Bar base excludes NFL seasonality.
SeasonalIndex	zone, month, index_value [TS]	120	10 zones x 12 months.
EvidenceStatus	zone, concept_type, status, demand_per_venue [TS]	80	Supply axis (§3.2). status in {Opportunity, Average, Saturated}.
EventMagnitude	event_class, magnitude	4	major_game 60, concert 35, festival 25, minor 15.
HolidayOccasion	occasion, date_rule, archetype, magnitude, primary_daypart	8	NEW. Calendar occasions; floating-date rules. magnitude in EventMagnitude units.
HolidayAffinity	occasion, concept_type, affinity [TS]	64	NEW. SIGNED -1..+1. Occasion-specific; do NOT substitute event_affinity.
Parameter	name, value	7	score_cap 150; proximity; weather bounds; utilization_default 1.0.
Operator	zone, concept_type, seats, staffing params, hours_overrides	per user	Staffing only, never demand. Not exported.

Band is a 6-row display reference (band, lower, upper, trend_up, trend_down), consumed as an option set plus the edge-trend rule in §3.1. **Export pipeline:** one CSV per table from the v71 workbook; quarterly Phase A rebuilds = re-export + re-import, no code changes.

3. The runtime calculation

Two independent outputs, kept separate by design: the DEMAND SCORE (how much appetite) and the EVIDENCE STATUS (whether that appetite is already served). The app displays both side by side; they must not be merged. Supply never enters

the demand score.

3.1 Demand score (the runtime formula)

```
base          = BaselineCurve.structural_base x SeasonalIndex[zone][month]
daypart_base  = base x BaselineCurve.daypart_fraction[daypart]

event_lift    = SUM events: EventMagnitude x ConceptType.event_affinity x proximity  (additive, >= 0)
holiday_lift  = HolidayOccasion.magnitude x HolidayAffinity[occasion][concept]      (SIGNED; 0 if none)
weather_fx    = 1 - weather_severity x ConceptType.weather_affinity

score = MAX( MIN( (daypart_base + event_lift + holiday_lift) x weather_fx , 150 ) , 0 )
```

Order of operations is validated: lifts (event + holiday) ADD, weather MULTIPLIES, the cap is applied, then the floor at 0. proximity = 1.0 inside the operator's zone, 0.5 adjacent, 0 otherwise (v1).

The floor at 0 is load-bearing. Holiday affinities are SIGNED and can be strongly negative (closure/suppression), so the adjusted score can be driven below 0 before the floor. Event affinities never do this (0..1, additive only).

CRITICAL — event_affinity is NOT holiday affinity. ConceptType.event_affinity is tuned for event crowds (games/concerts/festivals) and applies ONLY to EventMagnitude classes. Holidays use the separate, SIGNED HolidayAffinity table, whose profiles are often inverted relative to event_affinity. A stadium game lifts Sports Bar (1.0) and barely touches Fine Dining (0.15); Valentine's Day is the opposite (Fine Dining +1.0, Sports Bar -0.2). Applying the event column to holidays produces backwards results.

Holiday + event stacking (v1 decision). When a holiday and a real event coincide and describe the same demand (e.g. a sports bar on Thanksgiving Day with a Lions game in range), holiday_lift and event_lift BOTH apply additively in v1, with the cap absorbing the overlap. This may over-count shared demand; it is flagged for POS calibration and is the canonical Phase-2 revisit. Do not de-duplicate silently at runtime — keep both terms so the narration layer can attribute each.

Band mapping. 110-150 Exceptional · 85-109 Peak · 65-84 High · 40-64 Moderate · 20-39 Light · 0-19 Minimal.

Edge-trend display rule. Within 4 points of a band boundary, append the adjacent-band hint from Band.csv: a score of 82 (High, upper 84) displays “High — trending toward Peak”; a score of 87 (Peak, lower 85) displays “Peak — softening toward High”. Mid-band scores show the band name only. This keeps near-threshold output advisory rather than a hard flip.

Daypart split. daypart_score = score x BaselineCurve.daypart_fraction[daypart]. Fractions are zone-specific (encode the zone-daypart resolution). Dayparts: Morning 6-11, Midday 11-4, Dinner 4-9, Late Night 9-2.

3.2 Evidence status (the supply axis — lookup, not computed at runtime)

Each zone x concept carries a pre-computed EvidenceStatus from the workbook — a straight lookup, never recomputed at runtime. Display next to the demand band: Opportunity (high demand vs existing venues), Average (met by a moderate field), Saturated (well-served / contested). Example: “Fine Dining — Exceptional demand · Opportunity status.” The two axes are independent; never collapse them into a single label.

Staffing translation. covers = (daypart_score / 150) x seats x turn_rate x utilization; servers = CEILING(covers / covers_per_server); cooks = CEILING(covers / covers_per_cook); revenue = covers x avg_check. CEILING always rounds UP.

Utilization reconciliation: use utilization = 1.0 (Parameter.csv default). The turn rates already encode realistic throughput. Any 0.85 figure in earlier drafts is superseded — reconcile all staffing math to 1.0.

4. Bubble.io implementation map

ForeShift element	Bubble construct
Zones, ConceptTypes, curves, indices, evidence status, HolidayOccasion, HolidayAffinity, Band	Data types; bulk CSV import; privacy rules = no client access to [TS] fields
Bands, dayparts, event classes, statuses, occasions	Option sets
Demand-score formula	Backend workflow (server-side) computing score per day x daypart, including event_lift and holiday_lift, to a Forecast data type. Do NOT compute [TS] math in page expressions.

ForeShift element	Bubble construct
Holiday detection	Calendar resolver (date + floating-date rules from HolidayOccasion.date_rule). NOT an API call.
Evidence status	Lookup from EvidenceStatus by zone+concept; display alongside band. No runtime math.
Event detection API	API Connector to events API; query by zone geo_center+radius and date; classify by venue size/segment.
Weather API	API Connector to forecast API (NWS or OpenWeather); map to severity scale.
Narration layer	Server-side call to the AI narration service (\$5.5); receives score/band/trend/factors; returns advisory copy. Never alters scores.
Weekly refresh	Scheduled backend workflow per operator: resolve holidays, fetch events, fetch weather, recompute 7x4 Forecast records, request narration.
Operator onboarding	Form: zone dropdown (10), concept dropdown, operational params; hours pre-filled.
Display	Weekly 7x4 grid colored by band (with edge-trend label); evidence-status chip; staffing card; narration line per cell/day.
Quarterly update	Re-import CSVs from Excel master — no workflow changes.

5. Event, weather, and holiday ingestion

Implement the classification rules exactly.

Rule	Logic
Event magnitude	segment=Sports & capacity >= 30k -> major_game (60); Sports < 30k -> arena_game (45); Music >= 8k -> concert (35); capacity >= 3k -> festival (25); else minor (15)
Zone proximity	haversine(venue, zone centroid): <= 0.6 mi -> 1.0; <= 1.5 mi -> 0.5; else 0
Weather severity	thunder/snow OR precip >= 70% -> 0.5; rain OR precip >= 40% -> 0.25; sunny & 68-84F -> -0.10; else 0
Holiday detection	Resolve HolidayOccasion.date_rule against the calendar for the forecast date, including floating rules (4th Thu Nov, 2nd Sun May). Match -> apply holiday_lift via HolidayAffinity. No API.
Stacking	Same-day event + holiday: event_lift and holiday_lift both ADD, then weather multiplies, then cap 150, then floor 0 (see §3.1).

Coordinates. Zone centroids and venue coordinates MUST be verified against authoritative geodata; the proximity thresholds (0.6/1.5 mi) are calibration targets.

5.5 Narration layer (advisory copy — new)

The narration layer turns a computed forecast into operator-facing guidance. It is a server-side AI call that receives model outputs and returns copy. It narrates; it never computes or overrides a score.

Input contract. The narration call receives, per zone x concept x day(x daypart): score, band, edge-trend flag, the contributing factors (event(s), holiday, weather) with each factor's signed direction, and whether the lift is base-driven or overlay-driven (event/holiday). It must not receive raw [TS] coefficients beyond what is needed to name a factor.

Tone rules. (1) Advisory only — use “may”, “consider”, “based on early signals”; never issue directives or staffing commands. (2) Attribute — name the factor(s) driving the signal and their direction (“a major game nearby”, “Valentine's Day”, “forecast rain”). (3) Defer to the operator — frame extend/trim as something the operator confirms against the room.

Confidence modulation. Hedge harder when (a) the score is within 4 points of a band edge (use the trend flag — near a boundary is inherently less certain) or (b) the lift is overlay-driven rather than base-driven (a forecast about an external event is more falsifiable than a steady base pattern). Soften “likely” toward “may” in those cases.

Example output. “Early signals point to Peak demand tonight, trending toward Exceptional — driven by a major game nearby and clear weather. Based on your foot traffic, you may want to consider extending Late Night hours; confirm against what you see in the room.”

6. Custom-web alternative (non-Bubble path)

A single stateless scoring service (Node/Python, ~250 lines) exposing one endpoint (operator_id -> weekly forecast + evidence statuses + narration), backed by Postgres tables mirroring §2 (including HolidayOccasion/HolidayAffinity/Band), with cron jobs for weekly event/weather/holiday refresh and on-demand CSV import. The narration call is a server-side API request. No queue, cache, or ML infrastructure needed at launch.

7. Acceptance tests (validated v71 values — QA must reproduce to +/-0.2)

Values from the validated v71 workbook, month index 6 (June), normal weather unless noted. T16-T22 are the holiday-layer additions.

#	Scenario	Expected
T1	Campus Martius · Fine Dining · Sat · June · no events	145.0 -> EXCEPTIONAL
T2	Core City · Sports Bar · Tue · June · baseline	24.8 -> LIGHT
T3	T2 + major game (60 x 1.0 x 1.0)	84.8 -> HIGH
T4	T3 + storm (severity 0.5, affinity 0.30 -> x0.85)	72.1 -> HIGH
T5	Same game, Core City Fine Dining Tue (affinity 0.15)	base + 9 only — unrelated type must NOT spike
T6	Eastern Market · Fine Dining · Sat · June	61.6 -> MODERATE (uncapped)
T7	Mexicantown · Casual Dining · Sat · June	66.2 -> HIGH
T8	Mexicantown · Fine Dining · Sat · June	87.1 -> PEAK demand
T9	Evidence: Corktown · Fine Dining	Opportunity (high demand, ~0 venues)
T10	Evidence: Mexicantown · Casual Dining	Saturated (dense casual field)
T11	Two-axis: Corktown Fine Dining	Exceptional demand AND Opportunity, never merged
T12	Any score, any inputs	always within [0,150] (floor now real — see T17)
T13	Staffing: 2.3 servers required	rounds to 3 (CEILING, never down)
T14	Lions Sunday (Ford Field 65k) + rain 60% · Campus sports bar · Sep	base + 60 x1.0, x0.93 rain, cap -> 150 EXCEPTIONAL
T15	Same Lions game · New Center & Core City profiles	event lift = 0 (beyond 1.5 mi) — no geographic leakage
T16	Valentine's · Corktown Fine Dining Sat (holiday_lift 70 x +1.0)	125 + 70 -> cap 150 EXCEPTIONAL (lift on the right type)
T17	Christmas Day · Financial Coffee Thu (70 x -0.6 = -42)	68 - 42 -> 26 LIGHT; asserts suppression + floor at 0
T18	Thanksgiving Day · Campus Casual Thu (60 x -0.4)	83.6 - 24 -> 59.6 MODERATE (suppressed)
T19	Thanksgiving Day · Greektown Sports Bar Thu (60 x +0.3)	64.8 + 18 -> 82.8 HIGH (LIFTED — same day, opposite sign vs T18)
T20	Valentine's · Campus Sports Bar Sat (70 x -0.2)	109.6 - 14 -> 95.6 PEAK (inversion: NOT lifted)
T21	Stacking: Thanksgiving Day + Lions game · sports bar	holiday_lift +18 AND event_lift +60 both apply, cap absorbs
T22	Edge-trend display: score 82.8 (High, upper 84)	label = "High — trending toward Peak"

Evidence-status thresholds (reference): demand-per-venue = avg daily demand / (venue count + 0.5): Saturated < 7, Average 7-18, Opportunity > 18. Uniform across concepts. The app consumes the status label, not the threshold math.

8. Security, IP, and sequencing

Coefficients, curve values, daypart fractions, demand-per-venue figures, and the signed HolidayAffinity matrix are the trade-secret core: enforce server-side-only access (privacy rules on every [TS] field), exclude them from API responses and logs, and show operators bands and status labels rather than raw scores by default. The signed holiday affinity matrix is new trade-secret content in this release.

Sequencing gate. File the provisional patent BEFORE sharing this document or the data export with any external developer, and use NDAs regardless. The holiday layer is a structural change introducing new trade-secret content — route it through structural

review and the provisional filing before developer hand-off. Event/weather API terms of service must be reviewed for commercial-use compliance before launch.

9. Phasing

Phase	Scope
MVP (Tier 1, \$99)	Onboarding (10 zones); weekly 7x4 band grid with edge-trend labels; evidence-status chips (demand x opportunity); event intelligence layer; holiday/occasion layer (magnitudes flagged as uncalibrated research priors); narration layer (advisory copy); basic staffing card.
Phase 2 (Tier 2, \$199)	Weather layer live; staffing schedule builder; POS CSV upload -> calibration blend (incl. holiday magnitudes); daypart-windowed events and holidays; holiday+event de-duplication.

Note on MVP scope: the holiday layer is folded into MVP (not Phase 2) because holidays are a primary external factor for the start-of-day advisory — the demand signal would feel broken if it went silent on Thanksgiving or Valentine's. Magnitudes ship as research priors and calibrate against POS in Phase 2.